EDUCATION GUIDE

Education Guide

Education is part of every iGEM project, and most of the work is deciding what to teach, to whom, and how to know it worked. This guide turns what we learned from AeroSense's education work into steps another team can follow. It rests on our own experience and on one public survey, so treat it as a starting point and adapt it to your audience.

1. Find out what people already believe

Run a short survey before you design anything. Ours showed that people understood the risk they can see: most knew that removing visible mold does not make food safe. They understood the risks they cannot see much less well, and knowing the facts did not reliably change what they did. Only 58% understood that a normal look and smell does not mean no risk, and even respondents with higher knowledge often did not record when a pack was opened. A lecture saying "mold is dangerous" would have told them what many already knew.

- Put the exact statements you plan to teach into the survey, word for word, so you can ask them again later.
- Keep it short, and end with one open question.
- If your form is bilingual, make each answer option a single string. Google Forms can treat the same answer with and without a translated label as two categories and undercount both.

2. Choose one claim

Write the one sentence you want your audience to change a belief or habit about. Every activity should serve it. Ours was that heat and our senses cannot tell us whether food is safe.

3. Match the format to the audience

We used three layers, and you can reuse them:

- **Reach.** A survey, a game, and social media find many people briefly.
- **Structured learning.** A school camp, a workshop, and an online course give a smaller group enough time to understand.
- **Outside validation.** A conference, a peer team, and a guest lecture test whether the work holds up in front of experts.

Start broad, then deepen. Do not expect depth from a reach channel.

4. Measure the claim the same way each time

Ask the same statements, worded identically, before and after an activity. Keep reach numbers such as likes and saves separate from learning: they show interest, not understanding.

5. Make it accessible

Use colorblind-safe palettes, alt text on every graphic, and an icon plus a text label wherever color carries meaning. Offer the material in the language your audience uses.

6. Work with the people who know the audience

Ask teachers and hosts how long the session is, what level to pitch at, and what the group already covered. We worked with a primary school, a university lab, and a college.

7. Protect participants

Collect responses anonymously. Get consent before publishing photos or children's drawings, and show first names or none at all.

8. Document it and hand it on

Write the plan so someone who has never met you can run it. Publish it in open formats and say what it contains. Our Teacher Toolkit does this for our elementary school camp: a 40-minute lesson with slides, a whole-class game and a worksheet, packaged so that a teacher can run it without us. Then ask a teacher outside your team to try it, and record what they change.

Checklist

- ☐ We asked people what they believe before designing.
- ☐ We can state our one claim in a sentence.
- ☐ Each activity has an audience, a time, and a reason.
- ☐ We ask the same statements before and after.
- ☐ Our materials are accessible.
- ☐ Participants consented, and their data is anonymous.
- ☐ Someone outside the team can run it from what we published.

Our materials

- Odor Pixel Suite, our game: https://pinche567.github.io/iGEM-2026-wiki/game/apps/wiki-client/dist/index.html
- AeroSense Learning Lab, our online curriculum: https://pinche567.github.io/iGEM-2026-wiki/learning-platform.html
- Teacher Toolkit, our lesson pack: https://drive.google.com/drive/folders/1G4Iqt3uOsPHzQC_8z2P0Od9eDQ6hMdFZ?usp=sharing